Jiajin Liu

+1-934-210-4672 | jiajinliu@nyu.edu | Homepage | Google Scholar | LinkedIn

Education

- **New York University** Sept 2022 - Dec 2027 (Expected)

  Ph.D. in Computer Science, Courant Institute of Mathematical Sciences *New York, NY, USA*

  ◦ Advisor: Prof. Qiaoyu Tan (since Feb 2025)

- **New York University** Sept 2020 - May 2022

  M.S. in Computer Engineering, Tandon School of Engineering *New York, NY, USA*

  ◦ 2-Year Merit Scholarship

  ◦ GPA: 3.87/4.00

- **Northeastern University** Sept 2016 - May 2020

  B.Eng. in Software Engineering, School of Software Engineering *Shenyang, China*

  ◦ Outstanding Undergraduate Project & Thesis (Top 3%)

  ◦ GPA: 3.76/5.00

Research Interests

My research interests lie in ***Agentic Reasoning***, ***Graph Learning***, and ***Multimodal LLMs***. My PhD work develops LLM-based agents that decide where to search, which evidence to incorporate, and how to reason over graph-structured data. I also develop multimodal graph learning methods that integrate language, vision, and graph structure.

Internship & Research Experience

- **Adobe Research, San Jose** May 2026 - Aug 2026

  Research Scientist Intern, Mentors: Dr. Eunyee Koh, Dr. Somdeb Sarkhel, Dr. Junda Wu

  ◦ Research Focus: Causal Reasoning, LLMs, Recommender Systems

  ◦ Causality-Aware Recommendation: Developed an LLM-based user-modeling framework that captures need-event dynamics from text and improves recommendation through shared user archetypes *[S.3]*.

  ◦ Causal Preference Reasoning: Developed a context-aware causal reasoning pipeline using LLM-based event/context extraction, predictive state representations, and sheaf-based consistency analysis.

- **Data Intelligence and Reasoning Lab, NYU** Feb 2025 - Present

  PhD Candidate, Advisor: Prof. Qiaoyu Tan

  ◦ Research Focus: Agentic Reasoning, RAG, Graph Learning, Multimodal LLMs

  ◦ Agentic Graph Learning: Developed a search-augmented agentic reasoning pipeline for graph learning tasks *[S.2]*, and contributed to RL for agentic graph reasoning *[P.4]* ***(ACL '26)***.

  ◦ Efficient Agentic RAG: Designed preview-guided evidence admission for search agents, improving answer accuracy while reducing full-document reads *[S.1]*.

  ◦ Multimodal Graph Learning: Benchmarked VLMs across multimodal graph learning paradigms *[P.1]****(CVPR '26)***; Developed LLM-based *[P.2]****(KDD '26)*** *[P.3]****(CVPR '26)*** and VLM-based *[P.5]****(EMNLP '26)*** frameworks for learning over graph-structured data.

- **Network System Lab, NYU** Sept 2022 - Feb 2025

  PhD Student, Advisor: Prof. Guyue Liu

  ◦ Research Focus: Security and Efficiency in Datacenter Network Systems

  ◦ Security Management in Enterprise Networks: Designed an asset-based quantitative risk metric for network outsourcing, and identified limitations of strawman baselines. *[P.7]* ***(NDSS '25)***; Resource Allocation in Radio Access Networks: Designed a QoS-aware scheduling algorithm for resource allocation in next-generation RAN, improving service satisfaction rate under tight latency constraints.

- **High Speed Network Lab, NYU** Jan 2021 - May 2022

  Graduate Research Assistant, Advisor: Prof. H. Jonathan Chao

  ◦ Research Focus: Programmable Packet Scheduling in Datacenter Networks

  ◦ Scalable and Accurate Packet Scheduling in Datacenters: Proposed a programmable packet scheduler with accurate sorting and large capacity with 100 Gbps throughput on FPGAs in datacenter networks *[P.8]* ***(NSDI '24)***; Analysis on PIFO-Approximate Scheduling Algorithms: Investigated the impact of scheduling inversion on flow delay and fairness in datacenter networks. *[P.9]* ***(Computer Networks '24)***

SELECTED RESEARCH PROJECTS

- **Efficient Retrieval-Augmented Reasoning for Search Agents**

  SEEDS: Preview-Guided Evidence Admission for Search Agents [S.1]

  ◦ Formulated evidence admission (which retrieved documents an agent should read, and how many) as a reasoning-state-conditioned decision in agentic search-augmented reasoning, where fixed top-k prefixes waste context on irrelevant pages or miss key evidence;

  ◦ Designed a preview-guided admission mechanism that evaluates lightweight document snippets in the agent's current reasoning context, enabling adaptive selection before full-document inspection;

  ◦ Evaluated across knowledge-intensive QA benchmarks with local-corpus and live-web search; improved accuracy by 2.5 points while cutting fetched pages by $\sim$70%, consistently across agent backbones.

- **LLM Agentic Reasoning over Native Graph Topology**

  GraphSearch: Agentic Search-Augmented Reasoning for Zero-Shot Graph Learning [S.2]

  ◦ Identified structure–semantic entanglement as a key factor underlying the failure of LLM agentic reasoning on graph tasks, explaining why structure-unaware agents misinterpret graph topology on graph learning tasks including node classification and link prediction;

  ◦ Designed a graph-aware query planner that separates *where to search* from *what to retrieve*, and a two-stage retriever with topology-grounded candidates and hybrid semantic–structural ranking;

  ◦ Achieved the best average rank among zero-shot methods on six datasets across two backbones (Qwen2.5-32B/72B), with 1.3–5.8× retrieval speedup over a structure-unaware agentic baseline; extended with GRPO, yielding 2.4–28.7 percentage-point gains over the training-free agent.

- **Multimodal Large Language Models for Graph Learning**

  GraphVLM: Benchmarking Vision Language Models for Multimodal Graph Learning [P.1]

  ◦ Developed the first systematic benchmark for multimodal graph learning with VLMs, unifying three functional roles of Encoder, Aligner, and Predictor across GNN-, LLM-, and VLM-based paradigms;

  ◦ Investigated both prompt-level augmentation and latent-space fusion strategies for modality–structure alignment under pretrained and supervised fine-tuning settings;

  ◦ Revealed that fine-tuned VLMs serve as the most effective backbone predictors, achieving the strongest and most consistent performance gains across six real-world graph datasets.

- **Causality-Aware Recommendation with LLM-based User Modeling**

  CAROUSEL: Causality-Aware Recommendation via Archetype Steering over LLM-Extracted Event-Level Markov Chains [S.3]

  ◦ Proposed text-based, causality-aware user modeling to capture how user needs evolve toward preferences and decisions, using these dynamics to improve personalized recommendation;

  ◦ Designed an LLM-based pipeline to extract need-events and construct per-user Markov chains, regularized sparse transitions through shared need-chain archetypes, and ranked items by combining need–item affinities with archetype-derived, goal-directed state values without retraining the backbone;

  ◦ Improved collaborative, causality-aware, and LLM-based recommenders via plug-in integration on benchmarks (+11.6% to +84.2% relative), with substantial cold-start and partial-conversation gains and interpretable user-need dynamics;

Publications and Preprints

P=PUBLICATION, S=UNDER REVIEW

[S.1] **SEEDS: Preview-Guided Evidence Admission for Search Agents**.
Jiajin Liu, *Chenxi Xu, Dongzhe Fan, Yuanfu Sun, Qiaoyu Tan. (Under review)*

[S.2] **GraphSearch: Agentic Search-Augmented Reasoning for Zero-Shot Graph Learning**.
Jiajin Liu, *Yuanfu Sun, Dongzhe Fan, Qiaoyu Tan. (Under review; accepted for poster presentation at the Stanford Data-Centric AI Workshop, 2026)*

[S.3] **CAROUSEL: Causality-Aware Recommendation via Archetype Steering over LLM-Extracted Event-Level Markov Chains**.
Jiajin Liu, *Junda Wu, Qiaoyu Tan, Somdeb Sarkhel, Eunyee Koh. (Under review)*

[S.4] **Does Latent Chain-of-Thought Actually Reason? A Single, Unifying Taxonomy for Latent Reasoning**.
Jingjie Qiu, Sheldon Yu, Xunyi Jiang, ***Jiajin Liu****, Zihan Huang, Zhendong Chu, Tong Yu, Julian McAuley, Junda Wu. (Under review)*

[P.1] **GraphVLM: Benchmarking Vision Language Models for Multimodal Graph Learning**.
Jiajin Liu, *Dongzhe Fan, Chuanhao Ji, Daochen Zha, Qiaoyu Tan.* In *The IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.* ***(CVPR '26 Main)***

[P.2] **MLaGA: Multimodal Large Language and Graph Assistant**.
Dongzhe Fan, ***Jiajin Liu****, Yi Fang, Djellel Difallah, Qiaoyu Tan. Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2026.* ***(KDD '26 Oral)***

[P.3] **Mario: Multimodal Graph Reasoning with Large Language Models**.
Yuanfu Sun, Kang Li, Pengkang Guo, ***Jiajin Liu****, Qiaoyu Tan.* In *The IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.* ***(CVPR '26 Main)***

[P.4] **AgentGL: Towards Agentic Graph Learning with LLMs via Reinforcement Learning**.
Yuanfu Sun, Kang Li, Dongzhe Fan, ***Jiajin Liu****, Qiaoyu Tan.* In *The 64th Annual Meeting of the Association for Computational Linguistics, 2026.* ***(ACL '26 Main)***

[P.5] **One Model, Many Graphs: Learning over Attributed Graphs across Heterogeneous Modalities with Vision-Language Models**.
Yifang Chen, Jiayi Yang, Yuanfu Sun, ***Jiajin Liu****, Qiaoyu Tan. In The 2026 Conference on Empirical Methods in Natural Language Processing.* ***(EMNLP '26 Main)***

[P.6] **GraphVerse: A Comprehensive Visual Graph Reasoning Benchmark for Multimodal Large Language Models**.
Yuanfu Sun, Yuanhang Ren, Kang Li, Chuanhao Ji, Jiaxi Li, ***Jiajin Liu****, Ninghao Liu, Qiaoyu Tan. In The 2026 Conference on Empirical Methods in Natural Language Processing.* ***(EMNLP '26 Main)***

[P.7] **Heimdall: Towards Risk-Aware Network Management Outsourcing**.
Yuejie Wang, Qiutong Men, Yongting Chen, ***Jiajin Liu****, Gengyu Chen, Ying Zhang, Guyue Liu, Vyas Sekar.* In *Proceedings of the 2025 Network and Distributed System Security Symposium, 2025.* ***(NDSS '25)***

[P.8] **Sifter: An Inversion-Free and Large-Capacity Programmable Packet Scheduler**.
Peixuan Gao, Anthony Dalleggio, ***Jiajin Liu****, Chen Peng, Yang Xu, H. Jonathan Chao.* In *Proceedings of the 21st USENIX Symposium on Networked Systems Design and Implementation, 2024.* ***(NSDI '24)***

[P.9] **Inversion impact of approximate PIFO to Start-Time Fair Queueing**.
Junda Song, ***Jiajin Liu****, Peixuan Gao, Guyue Liu, H. Jonathan Chao. Computer Networks: The International Journal of Computer and Telecommunications Networking, 2024.* ***(Computer Networks '24)***

Professional Activities and Skills

- **Reviewer:** ICLR '27, WSDM '27, CIKM '26, CIKM '25 , IEEE TKDE, ACM TKDD, ACM Computing Surveys
- **Guest Lecturer:** DATS-SHU 369 Machine Learning with Graphs (SP'26, SP'25)
- **Teaching Assistant:** CSCI-SHU 360 Machine Learning (FA'24), CSCI-SHU 2314 Discrete Mathematics (SP'24)
- **Programming Languages:** Python, C++, C, Java, SQL
- **Developer Tools/Systems:** PyTorch, TensorFlow, HuggingFace Transformers, vLLM, CUDA, Git, Linux, Jupyter

Last updated: Nov 2026.